COMPUTATIONAL PHYSICS

Lecture 1

* Dimensional Analysis and Limiting cases.
  * Motivation
  * Use in validating formulae.
  * Use in deriving formulae.
  * Problems.

DIMENSIONAL ANALYSIS

- This is a calculus based course, a large part of which deals with modeling physical systems mathematically and simulating them on the computer. It is essential to cross-check correctness of results based on common-sense and physical intuition. Some rather obvious errors can be avoided sometimes.

- Useful in obtaining qualitative answers regarding behaviour of a physical system.

- Roughly stated, any eqn. describing a physical quantity must be dimensionally consistent.

- Principle of homogeneity of dimensions → Dimensions must match on both sides of an eqn.

Examples:- Verify correctness of the following formulas

(i) $s = vt + \frac{1}{2} gt^2$, (ii) $s = vt^2 + \frac{1}{2} at^2$, (iii) $s = \frac{at^2}{\sin^{-1} v}$.

Soln:- Let $[x]$ denote dim. of any quantity x.

Note:- dimensions $\equiv$ units.

(i) $[s] = [vt] + \frac{1}{2} [gt^2]$

$\Rightarrow L = (LT^{-1})T + \frac{1}{2} (LT^{-2} T^2)$

$\Rightarrow L = L$, which is true.

(ii) $[LHS] = [s] = L$

$[RHS] = [vt^2] + \frac{1}{2} [at^2]$

$= (LT^{-1} T^2) + (LT^{-2} T^2)$

$= (LT) + (L)$, which is absurd.

(iii) $s = \frac{at^2}{\sin^{-1} v}$ $\longrightarrow$ this relation involves a trigonometric term.

Now, trig. terms are usually defined as ... $= \frac{\text{length}}{\text{length}}$

for example ... $= \frac{\text{perpendicular}}{\text{hypotenuse}}$.

So, argument of a trigonometric function is dimensionless.

Another way to see this is that they expandable in power series.

Eg. $\sin x = x - \frac{x^3}{3!} + \dots$ So, x has to be dimensionless.

Again, $s = \dfrac{at^2}{\sin^{-1}(v)}$

$\Rightarrow \sin^{-1} v = \dfrac{at^2}{s}$

$\Rightarrow v = \sin\left(\dfrac{at^2}{s}\right)$.

$[L.H.S] = LT^{-1}$

$[R.H.S] = \left[\dfrac{at^2}{s}\right] = \dfrac{LT^{-2}T^2}{L} = 1$

$\therefore [L.H.S] \neq [R.H.S]$

<u>RULES</u>:- (i) both sides of an eqn. have the same dim.

(ii) all terms in a sum must have the same dim.

(iii) arguments of exp, log, sin, cos are dimensionless.

<u>Example</u>:- $A = Be^{-C} + \dfrac{(D_1 + D_2)}{E} + F$.

What can be inferred about the quantities in the above eqn. from dimensional considerations?

<u>Soln</u>:- $[C]$ = dimensionless

$[A] = [B] = [F]$

$[D_1] = [D_2]$

$\left[\dfrac{D_1 + D_2}{E}\right] = [A]$

- Dimensional analysis can yield qualitative answers.

<u>Example</u>:- Consider the <u>Gaussian</u> integral :-

$$I = \int_{-\infty}^{+\infty} dx \; e^{-\alpha x^2}$$

<u>Soln</u>:- <u>Remark</u>:- Consider that the integral is a fn of α.

$I \equiv I(\alpha)$. So, α is the only <u>scale</u> in the problem.

<u>Dimensional analysis</u>:- $[I] = [x]$, since $e^{-\alpha x^2}$ is dimensionless.

Also, $[\alpha] = [x]^{-2}$

$\Rightarrow [\alpha^{-1/2}] = [x]$

$\therefore I \propto \alpha^{-1/2}$

$I_{exact} = \sqrt{\frac{\pi}{\alpha}}$.

<u>Question</u>:- Does this logic hold for generic limits

$$I = \int_a^b dx \; e^{-\alpha x^2}$$

<u>Limiting cases</u> :- Sometimes limiting cases have obvious solutions.

Consider the <u>$\alpha \to 0$</u> limit :- $I \to \infty$ in this limit

<u>$\alpha \to \infty$</u> limit :- $I \to 0$ " " ".

$I(\alpha) \propto \alpha^{-1/2}$ respects this.

For example:- $I(\alpha) \propto \alpha^{1/2}$ would <u>NOT</u>

- Dimensional analysis can be used to derive qualitative formula for physical problems.

<u>Example</u>:- (VERTICAL THROW) A ball is thrown upwards with speed v_0. Determine the time taken to fall back to earth.

<u>Soln</u>:- t = time taken to fall back to earth.

$$t = F(m, v_0, g)$$

$\Rightarrow t^{\alpha} m^{\beta} v_0^{\gamma} g^{\delta}$ should be dimensionless.

$$[t^{\alpha} m^{\beta} v_0^{\gamma} g^{\delta}] = T^{\alpha} M^{\beta} L^{\gamma} T^{-\gamma} L^{\delta} T^{-2\delta}$$

$$= T^{\alpha-\gamma-2\delta} M^{\beta} L^{\gamma+\delta}$$

$\gamma = \frac{1}{2}$

$\delta = -\frac{1}{2}$

$\therefore \beta = 0$

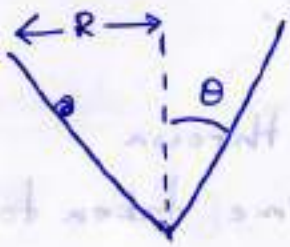
$$\left.\begin{aligned} \alpha - \gamma - 2\delta &= 0 \\ \gamma + \delta &= 0 \end{aligned}\right\}$$

$\gamma = -\delta$

$\gamma + 2\delta = \alpha$.

$$t \propto \sqrt{\frac{v_0}{g}}$$

Example :- (MARBLE ON INTERIOR OF CONE)



Time taken to complete one orbit?

Soln :- $t = f(\theta, m, R, g)$.

$$= h(\theta)\, m^{\alpha} R^{\beta} g^{\gamma}$$

$$[t] = M^{\alpha} L^{\beta} L^{\gamma} T^{-2\gamma}$$

$$\Rightarrow T = M^{\alpha} \cancel{L^{\beta-2\gamma}} L^{\beta+\gamma} T^{-2\gamma}$$

$\alpha = 0$

$\beta + \gamma = 0$

$-2\gamma = 1$

$\Rightarrow \gamma = -1/2$

$\beta = 1/2$

$$\therefore t \propto h(\theta)\sqrt{\frac{R}{g}}$$

PROB :- (PRESSURE OF A GAS CONFINED IN A BOX).

Soln :- Case 1 :- $p = m^{\alpha} v^{\beta}$

$$[p] = [m^{\alpha}][v^{\beta}]$$

$$\Rightarrow \left[\frac{F}{A}\right] = M^{\alpha} L^{\beta} T^{-\beta}$$

$$\Rightarrow \frac{MLT^{-2}}{L^2} = M^{\alpha} L^{\beta} T^{-\beta}$$

$$\Rightarrow ML^{-1}T^{-2} = M^{\alpha} L^{\beta} T^{-\beta}$$

$\alpha = 1$

$\left.\begin{array}{l}\beta = -1 \\ \beta = 2\end{array}\right\}$ absurd.

So, choice of variables is wrong.

Case 2 :- $p = m^{\alpha} v^{\beta} \left(\frac{N}{V}\right)^{\gamma}$

$[p] = [m^{\alpha}]\,[v^{\beta}]\,\left[\left(\frac{N}{V}\right)^{\gamma}\right]$

$\Rightarrow ML^{-1}T^{-2} = M^{\alpha} L^{\beta} T^{-\beta} L^{-3\gamma}$

$\Rightarrow ML^{-1}T^{-2} = M^{\alpha} L^{\beta-3\gamma} T^{-\beta}$

$\alpha = 1$

$\beta - 3\gamma = -1 \Rightarrow 3\gamma = 3 \Rightarrow \gamma = 1.$

$\beta = 2$

$\therefore p = m n v^{2}.$

$p_{exact} = \frac{1}{3} m n \bar{v}^{2}.$

Lecture 2

* Dimensional analysis continued.
* Function approximations.

COMBINING DIMENSIONAL ANALYSIS AND LIMITING CASES

Example:- (Hanging stretched cable)

Consider a cable/rope with a definite volume, instead of an idealised rope/spring (massless). ~~Suppose there is a weight suspended from~~ An idealised spring pulls across a point, whereas bulk materials pull across an area. Bulk modulus K of an isotropic material is defined as the proportionality const. between the change in material pressure Δp and relative increment in the material volume $\frac{(\Delta V)}{V}$ so that $\Delta p = -K\left(\frac{\Delta V}{V}\right)$.

Given a uniform, massive cable ~~of~~ hanging from one end, derive a formula for the stretch s as a fraction of unstretched length l.

Soln:- $s = f(l, \lambda, g, K)$.

Rewrite, $s = l\, f(\lambda, g, K)$

s	L
λ	ML^{-1}
l	L
g	LT^{-2}
K	$ML^{-1}T^{-2}$

$$\left[\frac{\lambda g}{l K}\right] = \frac{ML^{-1}\ LT^{-2}}{L\ ML^{-1}T^{-2}}$$

$$= \frac{MT^{-2}}{MT^{-2}} = 1$$

$\alpha = 1 \quad \beta = 1$

$\gamma = -1 \quad \delta = -1$

$$[\lambda^{\alpha} g^{\beta} l^{\gamma} K^{\delta}]$$

$$= M^{\alpha} L^{-\alpha} L^{\beta} T^{-2\beta} L^{\gamma} M^{\delta} L^{-\delta} T^{-2\delta}$$

$$= M^{\alpha+\delta} L^{-\alpha+\beta+\gamma-\delta} T^{-2\beta-2\delta}$$

$$\left.\begin{aligned} \alpha + \delta &= 0 \\ -\alpha + \beta + \gamma - \delta &= 0 \\ +2\beta + 2\delta &= 0 \end{aligned}\right\} \quad \begin{aligned} &\alpha = 1, \ \beta = 1 \\ &\gamma = -1, \ \delta = -1 \end{aligned}$$

$$\therefore \ s = l\, f\left(\frac{\lambda g}{l K}\right)$$

$$\Rightarrow \boxed{s = l\, f\left(\frac{\lambda g}{l K}\right)}$$

$f(x)$ can be :- (i) power law, ie, $f(x) \approx C x^{n}$

(ii) exponentially or log $f(x) \approx e^{x}$ or $\log x$.

as $x \gg 1$ or $x \ll 1$.

Physical considerations:- As $g \to$ large,

Material	K (Pascals)
Steel	10^{10}
Glass	10^{9}
Concrete	10^{9}.

- Power law asymptote:- $s = C \cdot l \left(\frac{\lambda g}{lK} \right)^n$

Note that $n > 0$. Otherwise, s would not increase with λ. Longer cables should also stretch more than shorter cables, so, $n < 1$.

$\therefore 0 < n < 1$.

Empirically:- $\boxed{0 < n < 1}$

Approximation of Functions

- Motivation :- Most dynamical systems are defined by change, which is usually represented by derivatives, ($\frac{dx}{dt}$, $\frac{dv}{dt}$ and so on), where x, v represent the variables needed to specify the system. Usually, they are obtained as solutions to differential eqns. of the form,

(i) $\frac{dy}{dx} = f(x)$. $\longrightarrow$ FIRST ORDER ODE

(ii) $\frac{d^2y}{dx^2} = f(x)$ $\longrightarrow$ SECOND ORDER ODE.

The first order ODE can ~~often~~ be written in the form,

$$\boxed{\begin{aligned} y_1 &= \frac{dy}{dx} \\ f(x) &= \frac{dy_1}{dx}. \end{aligned}}$$

$\Rightarrow$ Reduction to 1st order system.

$\Downarrow$

Solution can be obtained (in principle) by successive integration.

$$I = \int dx\, f(x).$$

Suppose $f(x)$ is a particularly difficult fn to evaluate / integrate.

- Can be use a simpler $f^{\underline{n}}$ as an approximation?

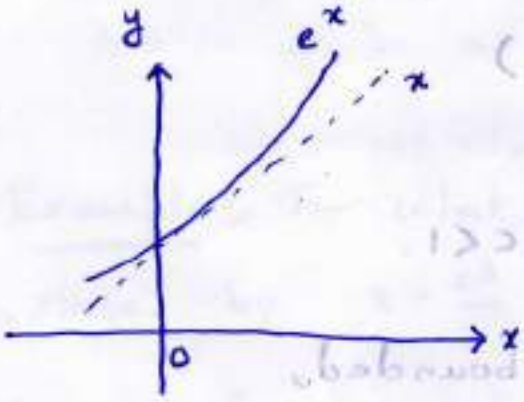
- Taylor polynomial.

Consider that ~~of fx~~ $f(x)$ has a Taylor series ~~of~~ the

Let $f(x)$ be approximated by a polynomial $p_n(x)$ of order n. in the neighbourhood of a point $x=a$.

For example :- a linear polynomial $p_1(x)$ has to satisfy,

$$\left.\begin{aligned} p_1(a) &= f(a) \\ p_2'(a) &= f'(a). \end{aligned}\right\} \Rightarrow p_1(x) = f(a) + (x-a) f'(a)$$



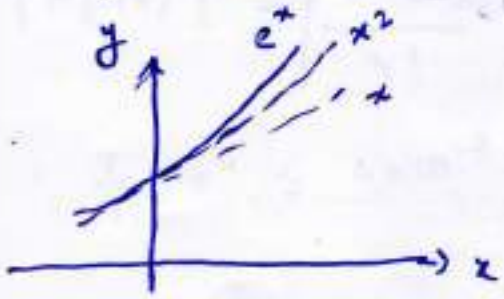
This is clearly not a good approximation as x increases.

$$p_2(x) = b_0 + b_1 x + b_2 x^2$$

The b's can be determined from

$$p_2(a) = f(a)$$
$$p_2'(a) = f'(a)$$
$$p_2''(a) = f''(a)$$

$$p_2(x) = f(a) + (x-a) f'(a) + \tfrac{1}{2}(x-a)^2 f''(a).$$



Taylor series :-

$$\boxed{p_n(x) = \sum_{j=0}^{n} \frac{(x-a)^j}{j!} f^{(j)}(a)}$$

ERROR IN TRUNCATING TAYLOR SERIES.

- REMAINDER THEOREM.

Assume that $f(x)$ has $(n+1)$ continuous derivatives in $\alpha \le x \le \beta$ and let $x=a$ belong to this interval. For Taylor polynomial $p_n(x)$, let $R_n(x) \equiv f(x) - p_n(x)$.

Then,

$$R_n(x) = \frac{(x-a)^{n+1}}{(n+1)!} f^{(n+1)}(c), \quad \text{where } a<c<x.$$

<u>Example:-</u> $f(x) = e^x$ and $a=0$. Put $x=1$.

$$R_n(x) = \frac{(x-a)^{n+1}}{(n+1)!} f^{(n+1)}(c)$$

$$R_n(1) = \frac{e^c}{(n+1)!}, \quad 0<c<1.$$

Since $e \approx 2.72 < 3$, $R_n(1)$ is bounded,

$$\frac{1}{(n+1)!} \le R_n(1) \le \frac{e}{(n+1)!} < \frac{3}{(n+1)!} \qquad e^0 \le e^c \le e^1$$

If we want $R_n(1) \le 10^{-9}$ (eg),

$$\boxed{\frac{3}{(n+1)!} \le 10^{-9}}$$

$\Rightarrow n \ge 12.$

Example:- $f(x) = \cos x$ at $x = 0.6$, when $a = 0$.
Estimate error.

Soln:- $f(x) = \cos x$ $\qquad R_n(x) = \dfrac{(x-a)^{n+1}}{(n+1)!} f^{(n+1)}(c)$.

$$p_2(x) = 1 - \frac{x^2}{2}$$

$$\cancel{R_n(x) = \frac{x^{n+1}}{(n+1)!}} \qquad R_2(x) = \frac{(x-0)^3}{3!} f^{(3)}(c).$$

$$\Rightarrow |R_2(x)|_{x=0.6} = \left| \frac{f^3(c)}{3!} x^3 \right| \Big|_{x=0.6}$$

$$= \frac{|\sin c|}{3!} |x|^3 \Big|_{x=0.6}$$

$$\leq \frac{|x|^3}{3!} = 0.036.$$

Example:- For what values of x can you replace $\sin(x)$ by $x - \frac{x^3}{6}$ with error no greater than 4×10^{-3}?

Soln:- $f(x) = \sin x$

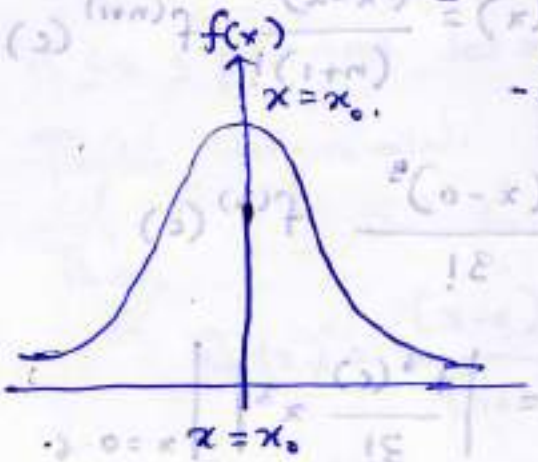
$$p_3(x) = x - \frac{x^3}{6}.$$

$$|R_3(x)| = \left| \frac{f^{(4)}(c)}{4!} x^4 \right| = \frac{|\sin(c)|}{4!} x^4 \leq \frac{1}{4} x^4.$$

$$\frac{1}{4!} x^4 \leq 4 \times 10^{-3} \Rightarrow |x| = 0.556.$$

- Application $\quad I = \int_{-\infty}^{+\infty} dx \; e^{-\alpha(x-x_0)^2}$

$$= \int_{-\infty}^{+\infty} dx \; e^{-(x-x_0)^2/\ell}.$$



α/ℓ controls the width of this graph, ie, it determines how sharply it drops off.

Integration over the entire range maybe computationally expensive. So, it might be feasible to integrate it to some finite value of x instead by expanding $f(x)$ in a Taylor series.

Remarks about remainder :-

$$f(x) = f(a) + f'(a)(x-a) + \cdots + \frac{f^{(n)}(a)}{n!}(x-a)^n + \frac{f^{(n+1)}(a)\,(x-a)^{n+1}}{(n+1)!} + \cdots$$

$$= p_n(x) + \underbrace{\frac{f^{(n+1)}(a)\,(x-a)^{n+1}}{(n+1)!} + \cdots}$$

$$\frac{f^{(n+1)}(\xi)\,(x-a)^{n+1}}{(n+1)!} = R_n(x) \rightarrow \text{captures the entire information after } p_n(x).$$

For reference,

$$e^x = 1 + x + \frac{x^2}{2!} + \dots + \frac{x^n}{n!} + \frac{x^{n+1}}{(n+1)!} e^c$$

$$\sin x = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \dots + (-1)^{n-1} \frac{x^{2n-1}}{(2n-1)!} + (-1)^n \frac{x^{2n+1}}{(2n+1)!} \cos c.$$

$$\cos x = 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \dots + (-1)^n \frac{x^{2n}}{(2n)!} + (-1)^{n+1} \frac{x^{2n+2}}{(2n+2)!} \cos c.$$

$$\frac{1}{1-x} = 1 + x + x^2 + \dots + x^n + \frac{x^{n+1}}{1-x} \quad ; x \neq 1$$

<u>Example</u> :- Approximate $\cos x$ in $-\pi/4 \leq x \leq \pi/4$ with an error $\leq 10^{-5}$. (Worked out in more detail later).

<u>Soln</u> :- $|R_n(x)| = \left| \frac{(x-a)^{n+1}}{(n+1)!} f^{(n+1)}(\xi) \right|$

Free to choose point to expand about, choose $a = 0$.

$$f(x) = \cos x = \sum_{n=0}^{\infty} (-1)^n \frac{x^{2n}}{(2n)!} \quad \text{(only even terms survive)}.$$

$$|R_{2n+1}(x)| \leq \left| \frac{x^{2(n+1)}}{(2n+2)!} \right| \leq 10^{-5}.$$

Worst case, $x = \pi/4$

$$\left| \frac{(\pi/4)^{2(n+1)}}{(2n+2)!} \right| \leq 10^{-5} \Rightarrow n \geq 8.$$

Example :- Approximate the integral,

$$I = \int_0^t \frac{dx}{1-x} = -\log(1-t)$$

Soln :- $$I = \int_0^t \frac{dx}{1-x} = \int_0^t dx\,(1-x)^{-1}$$

$$\Rightarrow -\log(1-t) = \int_0^t dx\,\left(1 + x + x^2 + \dots + x^n + x^{n+1} + \dots\right)$$

$$\Rightarrow -\log(1-t) = \int_0^t dx\left[\left(1 + x + x^2 + \dots + x^n\right) + \frac{x^{n+1}}{1-x}\right]$$

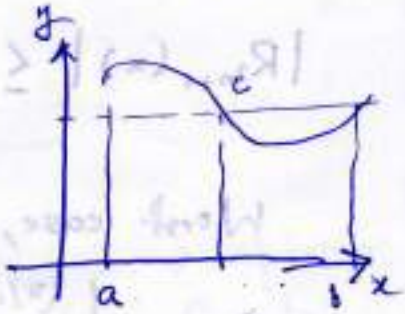
$$\Rightarrow -\log(1-t) = \int_0^t dx\left[\left(1 + x + x^2 + \dots + x^n\right)\right] + \underbrace{\int_0^t dt\,\frac{x^{n+1}}{1-x}\,dx}_{\text{Remainder term}}$$

(can be simplified by mean value th^m)

$$\int_0^t dx\,\frac{x^{n+1}}{1-x} = \frac{1}{1-c}\int_0^t dx\,x^{n+1} = \frac{1}{1-c}\,\frac{t^{n+2}}{n+2}, \quad 0 < c < t$$

INTEGRAL MEAN VALUE THEOREM :- If $w(x)$ is non-negative and integrable,

$$\int_a^b dx\,f(x)\,w(x)\,dx = f(\xi)\cdot\int_a^b dx\,w(x).$$



Example: Consider, $\sin t\, x = \frac{1}{x}\int_0^x dt\, \frac{\sin t}{t}$.

Derive Taylor polynomial approximation on $[-1,1]$ to above integral, such that max ~~error~~ error $\leq 5\times 10^{-9}$.

Soln: $f(x) = \sin x = x - \frac{x^3}{3!} + \cdots - (-1)^{n-1}\frac{x^{2n-1}}{(2n-1)!} + (-1)^n \frac{x^{2n+1}}{(2n+1)!}\cos c.$

$$\sin t\, x = \frac{1}{x}\int_0^x dt\left[1 - \frac{t^2}{3!} + \frac{t^4}{5!} + \cdots + (-1)^{n-1}\frac{t^{2n-2}}{(2n-1)!}\right]dt + R_{2n-2}(x)$$

$$= \frac{1}{x}\left[x - \frac{x^3}{3!\,3} + \frac{x^5}{5!\,5} + \cdots + (-1)^{n-1}\frac{t^{2n-1}}{(2n-1)!(2n-1)}\right] + R_{2n-2}(x)$$

$$R_{2n-2}(x) = \frac{1}{x}\int_0^x dt\,(-1)^n \frac{t^{2n}}{(2n+1)!}\cos(c)$$

$$|R_{2n-2}(x)| \leq \left|\frac{1}{x}\int_0^x dt\, \frac{t^{2n}}{(2n+1)!}\, dt\right|$$

$$= \left|\frac{x^{2n}}{(2n+1)\,(2n+1)!}\right| \leq 5\times 10^{-9}$$

Worst case $\Rightarrow x = 1$. $\Rightarrow \frac{1}{(2n+1)!\,(2n+1)} \leq 5\times 10^{-9}$

$\Rightarrow n \geq 5$.

Example:- Consider $I = \int_0^{1/3} dx\, e^{-x^2}$. Approximate this such that error $\leq 10^{-6}$

Soln:- $e^x = 1 + x + \frac{x^2}{2!} + \cdots + \frac{x^n}{n!} + \frac{x^{n+1}}{(n+1)!} e^{\xi}$.

$$e^{-x^2} = 1 - x^2 + \frac{x^4}{2!} + \cdots + (-1)^n \frac{x^{2n}}{n!} + (-1)^{n+1} \frac{x^{2(n+1)}}{(n+1)!} e^{\xi}.$$

$$\int_0^{1/3} dx\, e^{-x^2} = \int_0^{1/3} dx \left[\left(1 - x^2 + \frac{x^4}{2!} + \cdots + (-1)^n \frac{x^{2n}}{n!}\right) + (-1)^{n+1} \frac{x^{2(n+1)}}{(n+1)!} e^{\xi} \right]$$

$$\int_0^{1/3} \frac{x^{2(n+1)}}{(n+1)!} e^{\xi}\, dx \leq \frac{x^{(2n+3)}}{(n+1)!\,(2n+3)!} < 10^{-6}.$$

Worst case, $x = 1/3$ $\boxed{e^{1/3} < 1}$ After integrating, $\frac{(1/3)^{(2n+3)}}{(n+1)!\,(2n+3)!} \leq 10^{-6}$.

$n \geq 3$, inequality is satisfied.

Example:- (Repeated for clarity). Approximate $\cos x$ in $-\pi/4 \leq x \leq \pi/4$ with an error $\leq 10^{-5}$.

Soln:- $-\pi/4 \leq x \leq \pi/4$ is the given range. There is freedom to choose any arbitrary point in this region to expand $f(x)$ around. For convenience, choose $a = 0$.

$$f(x) = \cos x = f(a) + f'(a)(x-a) + \cdots + \frac{f^{(n)}(a)(x-a)^n}{n!}$$

$$+ \frac{f^{(n+1)}(a)(x-a)^{n+1}}{(n+1)!} + \cdots$$

$$= f(a) + f'(a)(x-a) + \cdots + \frac{f^{(n)}(a)(x-a)^n}{n!} + \frac{f^{(n+1)}(\xi)(x-a)^{n+1}}{(n+1)!}$$

$$= f(a) + f'(a)(x-a) + \cdots + \frac{f^{(n)}(a)(x-a)^n}{n!} + \underbrace{R_n(x)}_{\text{ambiguity in subscript?}}$$

Explicitly,

$$f(x) = \cos x = 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \frac{x^6}{6!} + \cdots$$

$$= \cancel{\sum_{n=0}^{\infty} (-1)^n \frac{x^{2n}}{(2n)!}} \quad \sum_{i=0}^{\infty} (-1)^i \frac{x^{2i}}{(2i)!}$$

$$= \cancel{\sum_{n=0}} \sum_{i=0}^{n} (-1)^i \frac{x^{2i}}{(2i)!} + (-1)^{n+1} \frac{x^{2(n+1)}}{(2n+2)!} f^{(n+1)}(\xi).$$

$f^{(n+1)}(\xi)$ will be either $\sin(\xi)$ or $\cos(\xi)$ $(0 \leq x < 1)$.

$$\therefore |R_{2n}(x)| \leq \underbrace{\left| \frac{x^{2(n+1)}}{(2n+2)!} \right|}_{\text{Increases with } x.} \leq 10^{-5}$$

So, worst case scenario is $x = \pi/4$.

$$\left| \frac{(\pi/4)^{2n+2}}{(2n+2)!} \right| \leq 10^{-5} \rightarrow \text{has to be solved for } n.$$

NUMERICAL DIFFERENTIATION

- From calculus, derivative of a function f at x_0 is defined by,

$$f'(x_0) = \lim_{h \to 0} \frac{f(x_0+h) - f(x_0)}{h}$$

- Truncation error shrinks as h decreases.

FORMULAS FOR DERIVATIVES.

1. Two-point formulas for 1st derivative.

(i) Backward difference:-

$$f(x_0-h) = f(x_0) - h f'(x_0) + \frac{h^2}{2} f''(\xi), \quad x_0-h \le \xi \le x_0.$$

$$\Rightarrow f'(x_0) = \frac{f(x_0) - f(x_0-h)}{h} + \frac{h}{2} f''(\xi)$$

(ii) Forward difference:-

$$f(x_0+h) = f(x_0) + h f'(x_0) + \frac{h^2}{2} f''(\xi), \quad x_0-h \le \xi \le x_0$$

$$\Rightarrow f'(x_0) = \frac{f(x_0+h) - f(x_0)}{h} - \frac{h}{2} f''(\xi)$$

2. Three-point formulas for 1st derivative.

$$f(x_0+h) = f(x_0) + h f'(x_0) + \frac{h^2}{2} f''(x_0) + \frac{h^3}{6} f'''(\xi_1)$$

$$f(x_0-h) = f(x_0) - h f'(x_0) + \frac{h^2}{2} f''(x_0) - \frac{h^3}{6} f'''(\xi_1).$$

$$\Rightarrow f'(x_0) = \frac{f(x_0+h) - f(x_0-h)}{2h} - \underbrace{\frac{h^2}{6} f''(\xi)}_{\text{using intermediate value theorem.}} \quad x_0 - h \le \xi \le x_0 + h$$

$\xi_1 \le \xi \le \xi_2$

Above formula takes into account a larger region while computing and is more accurate.

Higher order formulas can similarly be obtained using Taylor series.

$\boxed{\text{If } f(x) \text{ is smooth, order of error is dictated by power of } h.}$

APPLICATIONS IN SOLVING ODEs

- Initial value problem:-

$\frac{dy}{dx} = f(x)$. $\longrightarrow$ 1st order ODE, will need one condition.

Let $y(x = x_0) = y_0$ be this condition.

Now, discretise this equ. using the appropriate formula(s).

$$\frac{y(x+h) - y(x)}{h} = f(x)$$

$$\Rightarrow y(x+h) = y(x) + h f(x_0)$$

Let us ~~take th~~ say, that we interested in the solution from ~~x_0 to x_0+Nh~~. ~~and~~ $x = x_0 = a$ to $x = b$.

Divide the interval $[b, a]$ into n - subintervals, each of width $\frac{b-a}{n} = h$.

$$y(x+h) = y(x) + f(x).$$

EULER METHOD

Now, for $x = x_0$,

$$y(x_0+h) = \underbrace{y(x_0) + f(x_0)}_{\text{Completely known.}}$$

$$y(x_0+2h) = \underbrace{y(x_0+h)}_{\text{Found out earlier}} + \underbrace{f(x_0+2h)}_{\text{known.}}$$

$\vdots$

~~Example: (Particle moving under gravity).~~

3. Three-point formula for 2nd derivative.

$$f(x_0+h) = f(x_0) + h f'(x_0) + \frac{h^2}{2} f''(x_0) + \frac{h^3}{6} f'''(x_0) + \frac{h^4}{24} f^{(4)}(\xi_1)$$

$$f(x_0-h) = f(x_0) - h f'(x_0) + \frac{h^2}{2} f''(x_0) - \frac{h^3}{6} f'''(x_0) + \frac{h^4}{24} f^{(4)}(\xi_1)$$

Adding,

$$f(x_0+h) + f(x_0-h) = 2f(x_0) + h^2 f''(x_0) + \frac{h^4}{24} f^{(4)}(\xi)$$

$$\Rightarrow f''(x_0) = \frac{f(x_0+h) - 2f(x) + f(x_0-h)}{h^2} - \frac{h^2}{12} f^{(4)}(\xi)$$

Example :- (Particle moving under gravity).

A particle is thrown upwards with a velocity v_0. Find the solutions for the velocity and the position.

Soln :- $m\frac{dv}{dt} = -mg$ (upward motion).

$\Rightarrow \frac{dv}{dt} = -\frac{mg}{m}$, given, $v(t=0) = v_0$.

$x(t=0) = x_0$

$\Rightarrow \frac{dv}{dt} = -g$.

- The analytical solution is :-

$v = -\int dt\, g$

$\Rightarrow v = -gt + \text{const}$.

At $t=0$,

$v = v_0 \Rightarrow v_0 = \text{const}$.

$v = v_0 - gt$

$\Rightarrow \frac{dx}{dt} = v_0 - gt$.

$\Rightarrow dx = \int dt\,(v_0 - gt)$

$\Rightarrow x(t) = x_0 + v_0 t - \frac{1}{2}gt^2$.

Directly consider the 2nd order ODE,

$$\frac{d^2x}{dt^2} = -g$$

Option 1 :- Reduce this 2nd order ODE to a series of 1st order ODE's

Write $v = \frac{dx}{dt} \longrightarrow$ then solve this for x.

$$-g = \frac{dv}{dt} \longrightarrow \text{1st solve this eqn } v$$

Option 2 :- $\frac{d^2x}{dt^2} = -g$

Discretise using formula for 2nd order derivative.

$$\frac{x(t+h) - 2x(t) + x(t-h)}{h^2} = -g.$$

But, there will be a problem at $t=0$.

$$x(t_0+h) = 2x(t_0) - x(t_0-h) - h^2 g.$$

Sol/ Use initial condition $v = v_0$ at $t=0$.

$$\left.\frac{dx}{dt}\right|_{t_0} = v_0.$$

$$\Rightarrow \frac{-x(t_0-h) + x(t_0)}{h} = v_0. \quad \text{(Backward diff. formula)}.$$

$$\Rightarrow -x(t_0-h) = hv_0 - x(t_0)$$

$$\Rightarrow x(t_0-h) = x(t_0) - hv_0.$$

LECTURE 5

- Techniques to numerically evaluate integrals and derivatives set up.

- Motivations :- Use for modelling and simulation.
  - Set up a mathematical model to describe a given physical phenomenon and simulate on the computer

PROB :- (RADIOACTIVITY).

Atomic nuclei are composed of protons and neutrons, held together by nuclear forces. Stability based on whether the forces are balanced, ie, there is no "excess" of neutrons or protons. Unstable nuclei tend to decay to stable ones by emitting alpha, beta and gamma rays. These emissions can be measured and constitute 'radioactivity'.

Observation :- Radioactivity decreases with time exponentially. Can a mathematical model be set up to describe this phenomenon?

Considerations :- Radioactivity is actually quantum in nature and cannot be explained classically. Hence, it is not possible to take a really microscopic view of the situation. The mathematical model should suffice to explain observations.

Information:- Based on measurement of radioactivity, it is possible to construct a statistical model.

Assumptions:- Reasonable to assume that radioactivity will be proportional to total # of unstable nuclei. Let λ be the probability of decay of radionuclei per unit time.

$$\left[\text{Probability} = \frac{\#\text{ of favourable outcomes}}{\#\text{ of outcomes}}\right]$$

Hence, $dN = -\lambda N\, dt$.

$$\boxed{[\lambda] = T^{-1}}$$

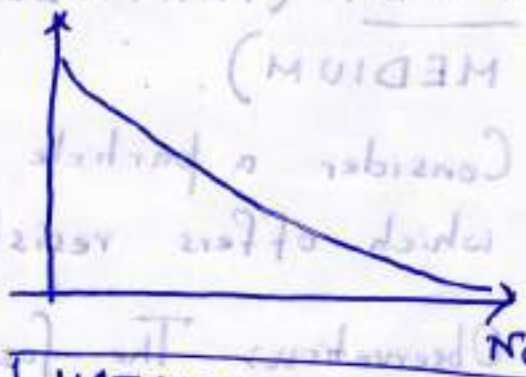
$$\Rightarrow \frac{dN}{N} = -\lambda\, dt$$

Integrating, $\ln N = -\lambda t + c_1$

$$\Rightarrow \ln N_0 = c_1$$

$$\therefore \ln \frac{N}{N_0} = e^{-\lambda t}$$

$$\Rightarrow N = N_0 e^{-\lambda t}.$$



MATCHES OBSERVATIONAL DATA

Physical interpretation of λ?

Consider avg/mean lifetime $\bar{T}$

$$\bar{T} = \frac{n_1 t_1 + n_2 t_2 + \cdots}{n_1 + n_2 + \cdots} = \frac{\sum n_i t_i}{\sum n_i} = \sum x_i t_i, \quad x_i = \frac{n_i}{\sum_i n_i}$$

Taking infinitesimal intervals, $\bar{T} = \int_0^\infty dt\, e^{-\lambda t} = \frac{1}{\lambda}$

Half-life:- Defn:- Time taken for half the nuclei to decay.

~~$T_{1/2} \rightarrow$ Note:- $t = \frac{1}{\lambda} \ln \frac{N_0}{N}$~~

~~$T_{1/2} = \frac{1}{\lambda} \ln \frac{1}{2}$~~

λ connected to how fast the sample decays.

$$N = N_0 e^{-\lambda t}$$

$$\frac{N_0}{2} = N_0 e^{-\lambda T_{1/2}}$$

$$\Rightarrow \ln\left(\frac{1}{2}\right) = -\lambda T_{1/2}$$

$$\Rightarrow \boxed{T_{1/2} = \frac{1}{\lambda} \ln 2.}$$

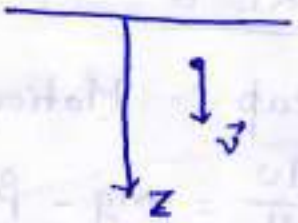
PROB:- (PARTICLE FALLING THROUGH A VISCOUS MEDIUM).

Consider a particle falling through a viscous medium which offers resistance to its motion.

Observation: The faster the particle moves, more is the viscous resistance.

Assumption:- Let us assume that the viscous force is proportional to the velocity of the particle. v.

The eqn. of motion is :-



$$m\frac{d\vec{v}}{dt} = (m\vec{g} - \beta\vec{v}).$$

All motion is along z-axis, so we drop the vector signs.

$$m\frac{dv}{dt} = mg - \beta v.$$

$$\Rightarrow \frac{dv}{mg-\beta v} = \frac{dt}{m}.$$

$$\Rightarrow \frac{-\beta\, dv}{mg-\beta v} = \left(-\frac{\beta}{m}\right)dt$$

$$\Rightarrow \ln(mg-\beta v) = -\frac{\beta}{m}t + c_1$$

At $t=0$, say, $v=v_0$.

$$\ln(mg-\beta v_0) = c_1.$$

$$\therefore \ln\frac{mg-\beta v}{mg-\beta v_0} = -\frac{\beta}{m}t.$$

$$\Rightarrow mg-\beta v = (mg-\beta v_0)e^{-\frac{\beta}{m}t}.$$

$$\Rightarrow \beta v = mg - (mg-\beta v_0)e^{-\beta t/m}$$

$$\Rightarrow v = \frac{mg}{\beta} - \left(\frac{mg}{\beta} - v_0\right)e^{-\beta t/m}.$$

$$\Rightarrow \boxed{v = \frac{mg}{\beta} + \left(v_0 - \frac{mg}{\beta}\right)e^{-\beta t/m}}$$

*

* Limiting case, :- $t\to\infty$.

$v(t\to\infty) = \frac{mg}{\beta}$. → Sensible?

Yes, $0 = mg - \beta v$

$\Rightarrow v = mg/\beta$

This const. vel. is termed terminal velocity

LECTURE 6

Recap :- Motion of particle with air resistance :-

$$m\frac{dv}{dt} = \underbrace{mg - \beta v}_{\text{Competing forces}}$$

When they balance, terminal velocity is reached.

Remark :- Why is the drag force prop. to v? What if we would like to incorporate terms higher order in v? Taylor expansion can be used as a systematic method to do this.

PROB :- (MOTION OF BICYCLE THROUGH RESISTIVE MEDIUM)

Consider a bicycle travelling against air resistance.

Observation :- Drag force increases with velocity.

Assumption :- Writing down $\frac{dv}{dt} = \frac{F}{m}$ is difficult, since there is no external force and the bicycle moves due to effort expended by the rider.

Let E = energy of bicyclet rider.

P = Power produced by rider. = const

Simplest model

$$\frac{dE}{dt} = P.$$

$$\Rightarrow \frac{d}{dt}\left(\frac{1}{2}mv^2\right) = P$$

$$\Rightarrow v\frac{dv}{dt} = \frac{P}{m}$$

$$\Rightarrow \boxed{\frac{dv}{dt} = \frac{P}{mv}} \Rightarrow \text{EOM.}$$

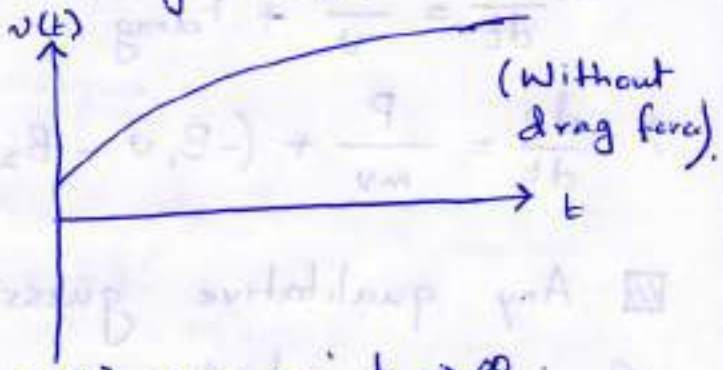
Assume:- Since P has been assumed to be constant, the EOM can be integrated,

$$\frac{v^2}{2} = \frac{Pt}{m} + C_1$$

When $v(0) = v_0$,

$$v^2 = v_0^2 + \frac{2Pt}{m}$$

$$\Rightarrow v = \sqrt{v_0^2 + \frac{2Pt}{m}}$$



$v(t) \to \infty$ as $t \to \infty$.

Observation:- Obviously, this is not what happens.

Discretise,

$$\frac{dv}{dt} = \frac{P}{mv}$$

$$\Rightarrow \frac{v_{i+1} - v_i}{\Delta t} = \frac{P}{mv_i}$$

$$\Rightarrow v_{i+1} = v_i + \frac{P}{mv_i}\Delta t$$

If $v_0 = 0$,

$$v_1 = \underbrace{0 + \frac{P}{m(0)}\Delta t}_{\text{blows up, unphysical, since } \frac{dv}{dt} \to \infty}$$

How to incorporate air resistance?

Let us assume a drag force of the form,

$$F_{drag} \equiv f(v)$$

$$= f(0) + f'(0)v + f''(0)v^2 + \cdots$$

$$= B_0 + B_1 v + B_2 v^2 + \cdots$$

Now, at $v = 0$, there is no drag force, so $B_0 = 0$.

$$F_{drag} = B_1 v + B_2 v^2$$

According to observation, $B_1 < 0$, $B_2 < 0$.

$$\therefore F_{drag} = \underbrace{-B_1 v}_{\text{dominates at low } v} \underbrace{- B_2 v^2}_{\text{dominates at high } v}$$

Resulting EOM :-

$$m\frac{dv}{dt} = \frac{P}{v} + F_{drag}$$

$$\Rightarrow \frac{dv}{dt} = \frac{P}{mv} + (-B_1 v - B_2 v^2)$$

Any qualitative guess regarding solution?

Consider a situation in which the velocity is low enough that $B_2 v^2$ can be neglected.

$$\boxed{\frac{dv}{dt} = \frac{P}{mv} - B_1 v} \Rightarrow \text{effective model.}$$

At small values of v; $\frac{P}{mv} >> B_1 v$.

$$\therefore \frac{dv}{dt} \approx \frac{P}{mv}.$$

$$\Rightarrow v = \sqrt{v_0^2 + \frac{2Pt}{m}}$$

At relatively larger values of v,

$$\frac{dv}{dt} \approx -B_1 v.$$

$$\Rightarrow v \approx e^{-B_1 t} + \text{const.}$$

Now include v^2 term and ignore v-term,

$$\frac{dv}{dt} = \frac{P}{mv} - B_2 v^2$$

Can the form of B_2 be guessed?

Consider the term, $B_2 v^2 \equiv f(\rho, A, v^2)$,

in other words, $B_2 \equiv B_2(\rho, A)$

<u>Option 1</u> :- Dimensional analysis :-

$$[F] = \rho^\alpha A^\beta (v^2)^\gamma$$

$$MLT^{-2} = (ML^{-3})^\alpha (L^2)^\beta (L^2T^{-2})^\gamma$$

$$= M^\alpha L^{2\beta - 3\alpha + 2\gamma} T^{-2\gamma}$$

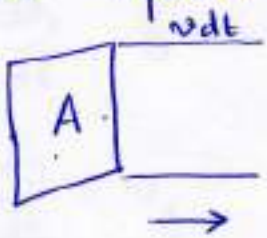
$$\left.\begin{aligned} \alpha &= 1 \\ 2\beta + 2\gamma - 3\alpha &= 1 \\ 2\gamma &= 2 \end{aligned}\right\} \Rightarrow \alpha = \beta = \gamma = 1.$$

$$\therefore F_{drag} \approx -\underbrace{C}_{\text{drag coefficient.}} \rho A v^2$$

<u>Option 2</u> :- "Exact" physical derivation.

As cyclist moves, air is pushed out of the way, work being done against the drag force.

Take an infinitesimal time interval dt, during which the cyclist moves distance $v\,dt$.



The mass of air moved is

$$m_{air} \approx \rho A v\, dt.$$

Assuming elastic collisions, the air molecules imparted velocity $-v$.

$\therefore E_{air} \sim m_{air} v^2/2$. = Work done by drag force.

$$\therefore F_{drag}\, v\, dt = E_{air} \Rightarrow F_{drag}\, v\, dt \approx -\rho A v\, dt\, \frac{v^2}{2}$$

$$\Rightarrow F_{drag} \approx -\tfrac{1}{2} \rho A v^2.$$

Caveat:- Consider the model when v is very very low.

As seen earlier, $\frac{dv}{dt} \to \infty$. What is going wrong?

Answer:- The assumption that power P is constant.

~~$\frac{P}{mv} \to$ if R also increases as v increases~~

At very low velocities, it may be more appropriate to solve the model,

$$\frac{dv}{dt} = \frac{F_0}{m} \rightarrow \text{this is regular at } v = 0.$$

$$\frac{dv}{dt} = \frac{P}{mv} - F_{drag} \rightarrow \text{can take over when the velocity is a little large}$$

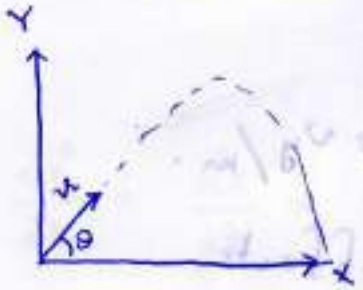
LECTURE 7

Recap:- Use of Taylor series as systematic approximation to a model

- The Taylor function is unknown, but coefficients deduced from physical considerations.
- This could also be done by fitting experimental data.

Motion in a plane.

- PROJECTILE MOTION (IDEALISED + REALISTIC).



Consider a particle thrown with velocity v_0, making angle θ with X-axis

The EOM are:-

$$\frac{d^2x}{dt^2} = 0 \quad , \quad \frac{d^2y}{dt^2} = -g.$$

Can be reduced to set of 1st order ODEs.

$$\left.\begin{aligned} \frac{dx}{dt} &= v_x \\ \frac{dv_x}{dt} &= 0 \\ \frac{dy}{dt} &= v_y \\ \frac{dv_y}{dt} &= -g \end{aligned}\right\}$$

$v_x = v_0 \cos\theta = \text{const.}$

$$x(t) = v_0 \cos\theta \, t$$

$$v_y(t) = v_0 \sin\theta - gt$$

$$y(t) = v_0 \sin\theta \, t - \tfrac{1}{2} g t^2.$$

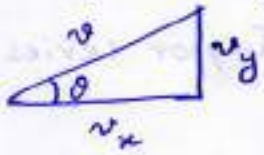
Time taken to reach max. height $= \dfrac{v_0 \sin\theta}{g}$.

Time taken to fall to ground $= (2v_0 \sin\theta / g)$ ~~$\frac{1}{2} g \sin\theta$~~

$$\text{Range} = x\left(t = \frac{2v_0 \sin\theta}{g}\right) = \frac{v_0 \cos\theta \, (2 v_0 \sin\theta)}{g} = \frac{v_0^2 \sin 2\theta}{g}$$

Now, incorporate air drag $F_{drag} = -B_2 v^2$ (say).

$v = \sqrt{v_x^2 + v_y^2}$.



$\cos\theta = \frac{v_x}{v}$

$\sin\theta = \frac{v_y}{v}$.

Resultant eqns are :-

$$\frac{dx}{dt} = v_x.$$

$$\frac{dv_x}{dt} = + F_{drag,x} = -\frac{B_2}{m} v^2 \frac{v_x}{v} = -B_2 v v_x/m$$

$$\frac{dy}{dt} = v_y$$

$$\frac{dv_y}{dt} = -g + F_{drag,y} = -g - B_2 v v_y/m.$$

— Now, try to incorporate other effects.

For instance, $B_2 \equiv B_2(\rho, A)$. Now, ρ changes with altitude.

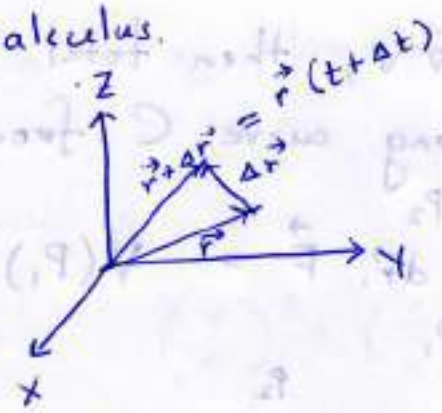
$$\rho = \rho_0 (e^{-y/y_0})$$

$$\therefore F^*_{drag} \equiv \frac{\rho}{\rho} F_{drag}(y=0)$$

Vector fields and calculus.

— $\vec{F} = m\dfrac{d\vec{v}}{dt}$.

$\vec{v} = \dfrac{d\vec{r}}{dt}$



Let $\vec{r} \equiv \vec{r}(t)$

$$\frac{\Delta\vec{r}}{\Delta t} = \frac{\vec{r}(t+\Delta t) - \vec{r}(t)}{\Delta t}$$

$$\Rightarrow \frac{\Delta\vec{r}}{\Delta t} = \frac{x(t+\Delta t) - x(t)}{\Delta t}\hat{x} + \frac{y(t+\Delta t) - y(t)}{\Delta t}\hat{y} + \frac{z(t+\Delta t) - z(t)}{\Delta t}\hat{z}$$

$$\frac{d\vec{r}}{dt} = \frac{dx}{dt}\hat{x} + \frac{dy}{dt}\hat{y} + \frac{dz}{dt}\hat{z}$$

<u>Work done</u> :- $W = \displaystyle\int_{\vec{r}_1}^{\vec{r}_2} d\vec{r}\cdot\vec{F}$ → conversion of potential to kinetic energy.

$$W = \frac{1}{2}m(v_2^2 - v_1^2)$$

<u>Conservative force field</u> :- A force/vector field $\vec{F}$ can be said to be conservative if there exists a scalar fn. V such that,

$$\vec{F} = -\vec{\nabla}V$$

$$\vec{\nabla}\times\vec{\nabla}V = \begin{vmatrix} \hat{x} & \hat{y} & \hat{z} \\ \partial_x & \partial_y & \partial_z \\ \partial_x V & \partial_y V & \partial_z V \end{vmatrix}$$

Equivalently, $\vec{\nabla}\times\vec{F} = 0$.

$= \hat{x}(0) = 0$

– If $F = -\vec{\nabla} V$, then total work done in moving the particle along curve C from P_1 to P_2 is,

$$W = \int_{P_1}^{P_2} d\vec{r} \cdot \vec{F} = V(P_1) - V(P_2)$$

* <u>Proof</u>:- $W = -\int_{P_1}^{P_2} d\vec{r} \cdot \vec{\nabla} V$ Total differential.

$$\Rightarrow W = -\int_{P_1}^{P_2} \left(dx \frac{\partial V}{\partial x} + dy \frac{\partial V}{\partial y} + dz \frac{\partial V}{\partial z} \right)$$

$$= -\int_{P_1}^{P_2} dV = V(P_1) - V(P_2)$$

$$\therefore \oint_C d\vec{r} \cdot \vec{F} = 0$$

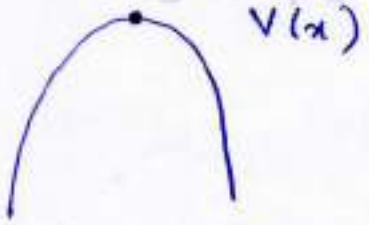
If $\vec{F}$ is not conservative, $W = \frac{1}{2} m (v_2^2 - v_1^2)$ would still hold for example, but relations such as conservation of energy would not.

<u>Equilibrium</u>:- $\cancel{\frac{\partial V}{\partial x}}$ $\vec{\nabla} V = 0$

$$\Rightarrow \frac{\partial V}{\partial x} = \frac{\partial V}{\partial y} = \frac{\partial V}{\partial z} = 0$$

Stable or unstable ?

STABLE



UNSTABLE.

Example:- Particle moves in the xy plane with position vector,

$$\vec{r} = a\cos\omega t\,\hat{x} + b\sin\omega t\,\hat{y}$$

$$x = a\cos\omega t \qquad \therefore \left(\frac{x}{a}\right)^2 + \left(\frac{y}{b}\right)^2 = 1$$

$$y = b\sin\omega t$$

$$\vec{F} = m\frac{d^2\vec{r}}{dt^2} = m\frac{d^2}{dt^2}\left[(a\cos\omega t)\hat{x} + (b\sin\omega t)\hat{y}\right]$$

$$= -m\left[-\omega^2 a\cos\omega t\,\hat{x} - \omega^2 b\sin\omega t\,\hat{y}\right]$$

$$= -m\omega^2\vec{r} \rightarrow \text{force always directed towards origin.}$$

↳ CENTRAL FORCE

Defn:- (i) Force always directed towards ~~and~~ or away from fixed point O.

(ii) Magnitude only depends on distance from O.

$$\vec{F} = f(r)\,\hat{r} = f(r)\frac{\vec{r}}{r}$$

EOM for particle in a central force field.

Theorem:- If a particle moves in a central force field, its path is a plane curve.

Proof:- $\vec{r}\times\vec{F} = f(r)\,\vec{r}\times\hat{r} = 0$

$$\Rightarrow \vec{r}\times\frac{d\vec{v}}{dt} = 0 \Rightarrow \frac{d}{dt}(\vec{r}\times\vec{v}) = 0 \Rightarrow \vec{r}\times\vec{v} = \vec{h}$$

$\Rightarrow \vec{r} \times \vec{v} = \vec{h}$

Identity, $\vec{r} \cdot (\vec{r} \times \vec{v}) = (\vec{r} \times \vec{r}) \cdot \vec{v} = 0$

$\cancel{\Rightarrow (\vec{r} \times \vec{v}) \cdot \vec{h} = \vec{h} \cdot \vec{h}}$

$\Rightarrow (\vec{r} \times \vec{v}) \cdot \vec{r} = \vec{h} \cdot \vec{r}$

$\Rightarrow \vec{h} \cdot \vec{r} = 0$, so motion takes place in a plane, since $\vec{h}$ is constant.

$m(\vec{r} \times \vec{v}) = m\vec{h} = \text{const}$ = Angular momentum conserved.

At any given t_0, consider plane P containing $\vec{r}_0$ and $\vec{v}_0$.

<u>Claim</u>:- $\vec{r}$ lies in P at all times.

Now, P is orthogonal to vector $\vec{n}_0 \equiv \vec{r}_0 \times \vec{v}_0$.

$\vec{r} \times \vec{v} = \vec{n}_0 = \vec{h} \quad \forall t$

$\vec{r}$ is orthogonal to $(\vec{r} \times \vec{v})$, $\vec{r}$ is orthogonal to $\vec{h}/\vec{n}_0 \ \forall t$.

LECTURE 15

VII FIRST-ORDER AUTONOMOUS SYSTEMS.

— $\frac{dx}{dt} \propto x$

$\Rightarrow \frac{dx}{dt} = \pm ax.$

Motivation:- Population model.
Let $x(t)$ denote the population of given species at time t and let $r(t,x)$ denote the diff. between birth and death rate.

If this population is isolated, $\frac{dx}{dt} = r(t,x)\,x(t)$.

For a simple model, assume, $r(t,x) = a = \text{const}$.

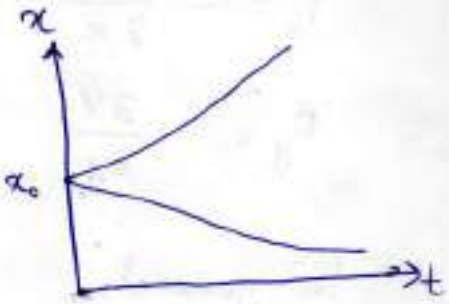
$$\boxed{\frac{dx}{dt} = \pm ax.}$$

$[a] = [T]^{-1}$

$\Rightarrow [at] \rightarrow$ dimensionless.

$\Rightarrow \frac{dx}{d(at)} = \pm x.$

$\Rightarrow \int \frac{dx}{x} = \pm \int dT$, where $T = at$.



$\Rightarrow \ln x = \pm T + c$

$\Rightarrow x = Ae^{\pm T}$

At $t = 0$, $x = x_0$. $\therefore x = x_0 e^{\pm T}$

$\Rightarrow \left(\frac{x}{x_0}\right) = e^{\pm T}$

$\Rightarrow \underbrace{X = e^{\pm T}}_{\text{Dimensionless.}}$

— Consider a more complicated system,

$\frac{dx}{dt} = a \pm bx.$

Case I :- $\frac{dx}{dt} = a - bx = b\left(\frac{a}{b} - x\right)$.

$\Rightarrow \frac{dx}{d(bt)} = \left(\frac{a}{b} - x\right)$

$[b] = [T]^{-1}$

$\Rightarrow [bT] \rightarrow$ dimensionless.

$T = bt$. $\quad x_0 = \frac{a}{b}$.

$\frac{dx}{dT} = (x_0 - x) = x_0\left(1 - \frac{x}{x_0}\right)$.

$\Rightarrow \frac{d(x/x_0)}{dT} = \left(1 - \frac{x}{x_0}\right)$

$\Rightarrow \frac{dX}{dT} = (1 - X)$.

$\Rightarrow -\ln(1-X) = T + c$

$\Rightarrow (1-X) = A e^{-T}$.

At $t = 0 \Rightarrow T = 0$, $x = 0 \Rightarrow X = 0$

$(1 - 0) = A$

$\Rightarrow A = 1$

$\therefore (1 - X) = e^{-T}$

$\Rightarrow \underbrace{X = (1 - e^{-T})}_{\text{Dimensionless}}$.

In dimensionfull terms, the soln is,

$x = x_0(1 - e^{-bt})$

$= \frac{a}{b}(1 - e^{-bt})$.

Small-time limit ($T \ll 1$)

$e^{-T} \simeq 1 - T + \ldots$

$X \simeq (1 - 1 + T) = T$.

$\therefore x = at$

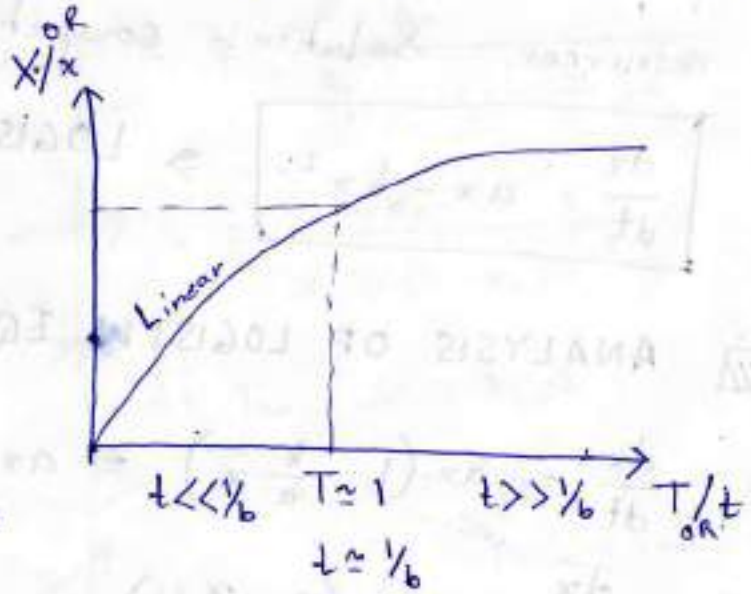
Large-time limit ($T \gg 1$),

$e^{-T} \rightarrow 0$.

$X \simeq 1 \Rightarrow x \simeq x_0 = a/b$.

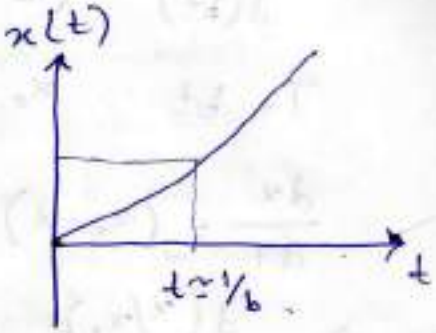
$\boxed{t \simeq 1/b}$ $\rightarrow$ point where the behaviour of the soln changes

$\rightarrow$ depends on parameter b of system.



At $t \simeq 1/b$, $x \simeq 0.63 x_0$. $\Rightarrow x = x_0(1 - e^{-1})$.

– "Non-dimensionalizing" the eqn. makes it easier to identify ~~scale~~ natural scale of the system.



Case II :- $\frac{dx}{dt} = a + bx$.

$$x(t) = \frac{a}{-b}\left(1 - e^{bt}\right) = \frac{a}{b}\left(e^{bt} - 1\right).$$

When $t \ll 1/b$, $\quad x \simeq \frac{a}{b}(bt) \simeq at$.

When $t \gg 1/b$, $\quad x(t) \simeq \frac{a}{b} e^{bt}$.

PROBLEMS WITH SIMPLE MODEL FOR POPULATION GROWTH.

$x(t) = x_0 e^{at}$ → exponential growth, solution blows up.

Linear models for population growth are suitable as long as population is not too large. But for very large population, members are competing with each other for resources. Solution :- competition term,

$$\boxed{\frac{dx}{dt} = ax - bx^2.} \Rightarrow \text{LOGISTIC EQUATION}$$

ANALYSIS OF LOGISTIC EQUATION.

$$\frac{dx}{dt} = ax\left(1 - \frac{b}{a}x\right) = ax\left(1 - \frac{x}{a/b}\right).$$

$$\Rightarrow \frac{dx}{d(at)} = x\left(1 - \frac{x}{a/b}\right).$$

Define $K = (a/b)$.

Define $T = at$, $X = \frac{x}{K}$.

$$\Rightarrow \frac{dx}{d(at)} = x\left(1 - x/K\right)$$

$$\therefore \frac{dx}{dT} = x(1-X) = K\frac{x}{K}(1-X)$$

$$\Rightarrow \frac{dX}{dT} = X(1-X).$$

$$\Rightarrow \frac{dX}{X(1-X)} = dT$$

Let $\frac{1}{X(1-X)} = \frac{A}{X} + \frac{B}{1-X}$ Solving, $A=1$, $B=1$.

$$= \frac{A+(B-A)X}{X(1-X)}$$

$$\int\left(\frac{dX}{X} + \frac{dX}{1-X}\right) = T + c$$

$$\Rightarrow \ln X - \ln(1-X) = T + c$$

$$\Rightarrow \ln\frac{X}{1-X} = T + c.$$

$$\Rightarrow \frac{X}{1-X} = A e^{T}$$

$$\Rightarrow X = A e^{T}(1-X)$$

$$\Rightarrow X(1+Ae^{T}) = Ae^{T}$$

$$\Rightarrow X(T) = \frac{Ae^{T}}{1+Ae^{T}}$$

$$\Rightarrow X = \frac{1}{1+A^{-1}e^{-T}}$$

$$\Rightarrow X = \frac{1}{1+\frac{(1-X_0)}{X_0}e^{-T}}$$

$$\Rightarrow X = \frac{X_0}{X_0+(1-X_0)e^{-T}}$$

At $t=0$, $X = X_0$ $(x = x_0)$.

$$\frac{X_0}{1-X_0} = A$$

$$\cancel{\Rightarrow X_0 = A - AX_0}$$

$$\cancel{x = \frac{K^2 x_0}{Kx_0 + (1/K x_0)e^{-Kt}}}$$

$$x = \frac{x_0}{\frac{x_0}{K} + \left(1-\frac{x_0}{K}\right)e^{-T}}$$

$$\Rightarrow x = \frac{Kx_0}{x_0 + (K-x_0)e^{-T}}$$

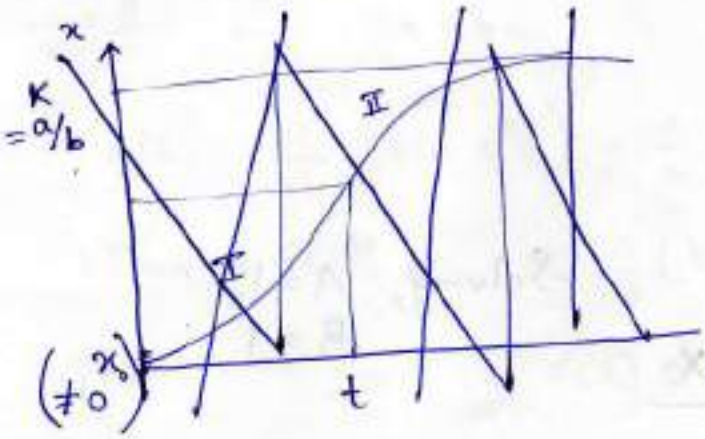
At large time limit,

$x \to K = a/b$ (for any initial value x_0).

At small-time limit,

$$x \simeq \frac{Kx_0}{x_0(1-e^{-T}) + Ke^{-T}}$$

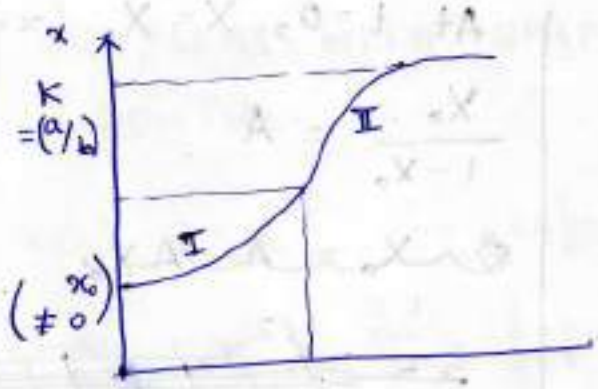
$$\Rightarrow \quad x \simeq \frac{x_0 e^{at}}{1 + \frac{x_0}{K} at} \longrightarrow \text{dominates time evolution}$$



I → early-time exponential growth

II → growth at decreasing rate at later time.

Growth saturates at $K = (a/b)$ irrespective of initial x_0. This is termed as the carrying capacity.



$$\frac{dX}{dT} = F(X) = X - X^2$$

$$\frac{d^2X}{dT^2} = \frac{dF}{dX}\frac{dX}{dT} = F\frac{dF}{dX} \Rightarrow \frac{d^2X}{dT^2} = 0 \text{ when } \frac{dF}{dX} = 0$$

$$\frac{dF}{dX} = 1 - 2X = 0 \Rightarrow X = \frac{1}{2}$$

$\frac{d^2X}{dT^2}$ changes sign at $X = \frac{1}{2}$ $\Rightarrow$ $F(X)$ has turning point

When $X > \frac{1}{2}$, $\frac{d^2X}{dT^2} < 0$, so growth occurs at decreasing rate

When $X < \frac{1}{2}$, $\frac{d^2X}{dT^2} > 0$, so growth occurs at increasing rate.

Physical interpretation :- At $X = 1/2$, the non-linear term $(-bx^2)$ begins to affect.

$$X = \frac{1}{2} = \frac{1}{1 + A^{-1} e^{-T_{n1}}} \Rightarrow 2 = 1 + A^{-1} e^{-T_{n1}}$$

$$\Rightarrow T_{n1} = \ln\left(\frac{1}{A}\right)$$

$$= \ln\left(\frac{1 - X_0}{X_0}\right).$$

$$\Rightarrow a t_{n1} = \ln\left(\frac{K - x_0}{x_0}\right)$$

$$\Rightarrow t_{n1} = \frac{1}{a} \ln\left(\frac{K}{x_0} - 1\right).$$

Constraint, $t_{n1} > 0$.

$$\frac{K}{x_0} - 1 > 1.$$

$$\Rightarrow x_0 < K/2$$

$$\Rightarrow \underbrace{X_0 < 1/2}_{\text{recall that this the turning point}}.$$

Hence if $\frac{K}{2} < x_0 < K$, growth occurs at decreasing rate only.

WAYS TO GUESS QUALITATIVE FEATURES OF FIRST-ORDER AUTONOMOUS ODE ?

$\dot{x} = f(x)$ → general form.

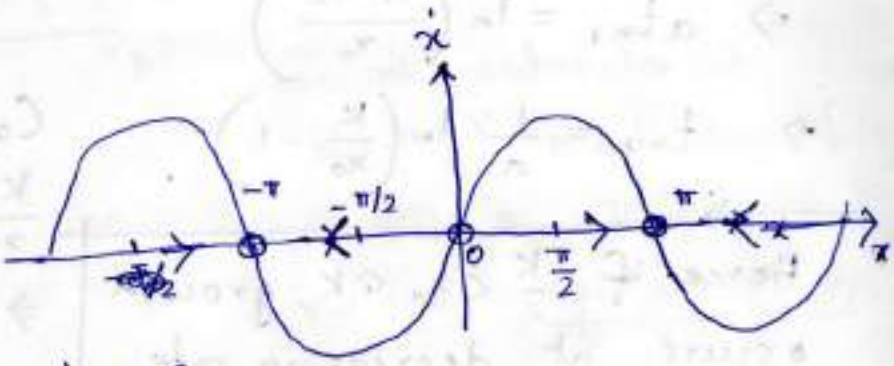
- Visualise t as time, x as the position of a particle moving along x-axis and $\dot{x}$ is the velocity of same.
- ~~To specify~~ When $\dot{x} = 0$, such points are termed "FIXED POINTS".
- There are two types of fixed points – stable and unstable. / (attractors and repellers)

Example:- Consider the logistic eqn:-

$$\dot{x} = ax - bx^2 = x(a - bx).$$

The fixed points are $x^* = 0, \left(\frac{a}{b}\right)$. Has been observed that at long-times, solutions tend towards (a/b) for given initial values.

Example:- $\dot{x} = \sin x$.



$$\Rightarrow \frac{dx}{dt} = \sin x$$

$$\Rightarrow \frac{dx}{\sin x} = dt$$

$$\Rightarrow t = -\ln|\operatorname{cosec} x + \cot x| + C.$$

Say, $x = x_0$ at $t = 0$,

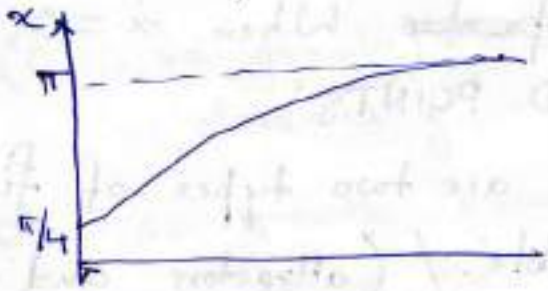
$$t = \ln\left|\frac{\operatorname{cosec} x_0 + \cot x_0}{\operatorname{cosec} x + \cot x}\right|$$

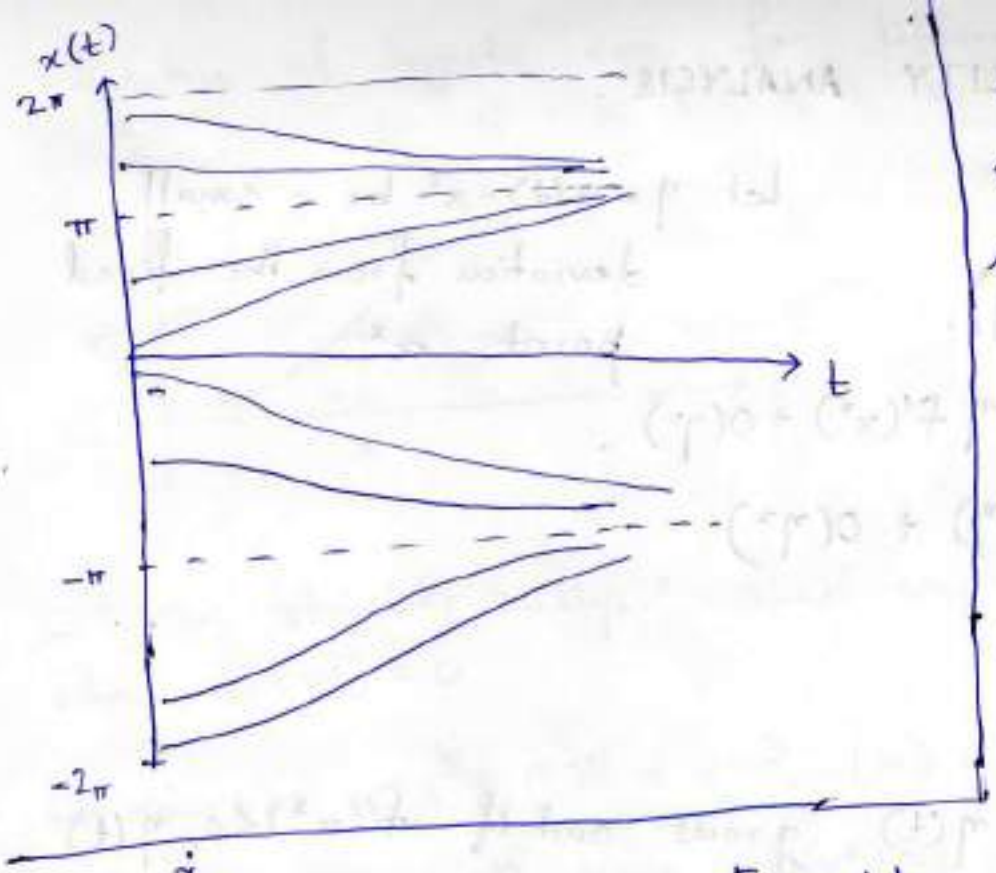
Fixed points are given by,

$$\sin x^* = 0$$

$$\Rightarrow x^* = 0, \pm\pi, \ldots$$

Reminder:- visualise particle moving along x-axis at velocity $\dot{x}$ at $t = 0$. If $\dot{x} > 0$, it means velocity of particle is increasing with x and vice versa. If the particle tends to move back towards a fixed point, then the fixed points is <u>stable</u>, else it is unstable.





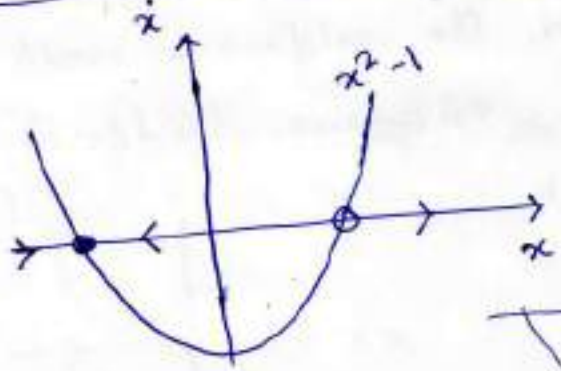
Fixed points represent equilibrium solutions.

<u>Example</u>:- $f(x) = x^2 - 1$

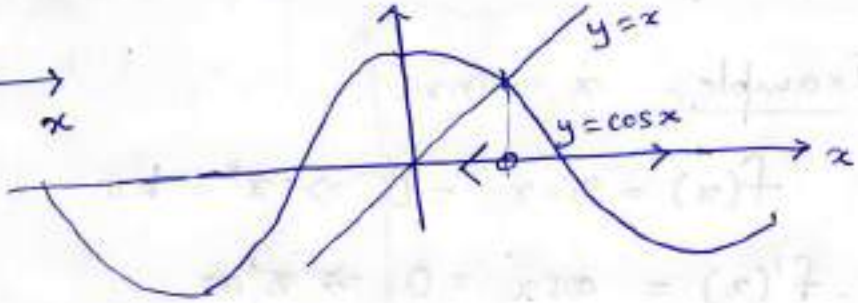
$f(x) = x(x-1)$

$f(x^*) = 0$

$\Rightarrow x^* = 0, 1$



<u>Example</u>:- $f(x) = x - \cos x$.



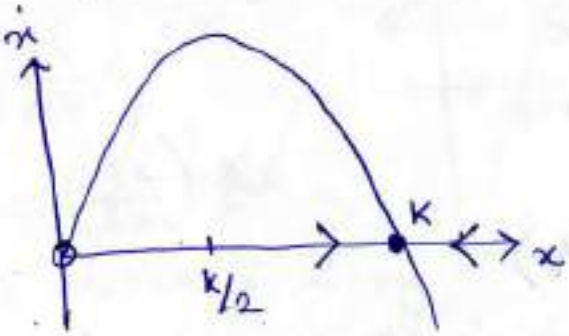
<u>Example</u>:- Logistic eqn.. $\dot{x} = ax - bx^2$

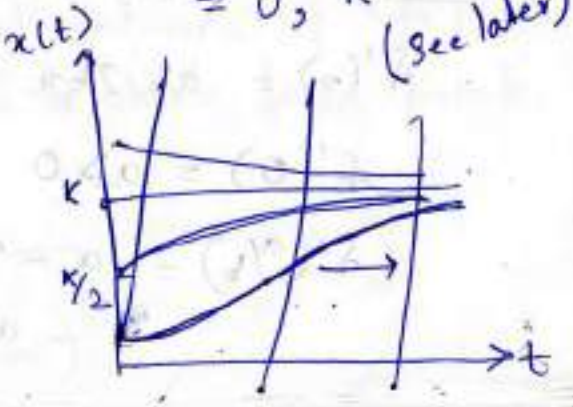
<u>OR</u> $\dot{X} = X - X^2$.

Fixed points are $x^*(a - bx^*) = 0 \Rightarrow x^* = 0, (a/b)$.

$= 0, K$

(see later)





LINEAR STABILITY ANALYSIS.

$\dot{x} = f(x)$

Let $\eta = x(t) - x^*$ be a small deviation from the fixed point x^*.

$\Rightarrow \dot{\eta} = f(x^* + \eta)$.

$\Rightarrow \dot{\eta} = f(x^*) + \eta f'(x^*) + O(\eta^2)$.

$= \eta f'(x^*) + O(\eta^2)$

Linearise,

$\dot{\eta} = \eta f'(x^*)$.

If $f'(x^*) > 0$, $\eta(t)$ grows and if $f'(x^*) < 0$, $\eta(t)$ decays.

<u>Example</u>:- $\dot{x} = \sin x$.

$f(x) = \sin x = 0 \Rightarrow x^* = k\pi$

$f'(x) = \cos x = 0$

$\Rightarrow f'(x^*) = \cos k\pi = 0 \begin{cases} 1, & k \text{ even} \\ -1, & k \text{ odd}. \end{cases}$

$\Rightarrow$ x^* is unstable if k is even and stable if k is odd.

<u>Example</u>:- $f(x) = ax - bx^2$

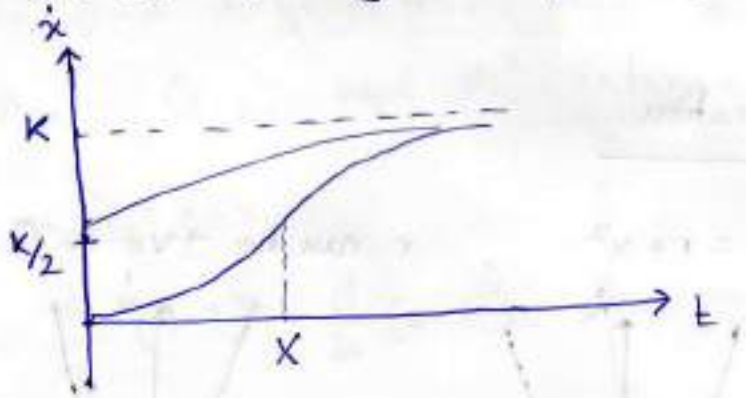
$f'(x) = a - 2bx$

$f'(0) = a > 0$

$f'(a/b) = a - 2(b)(a/b)$

$= -a < 0$

Solution of logistic eqn for different initial conditions.



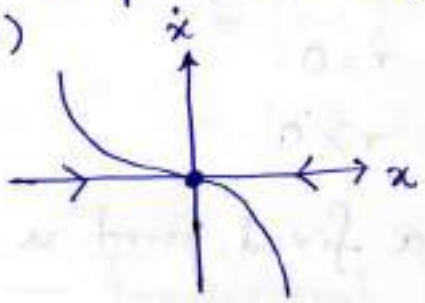
– Linear stability analysis cannot say anything conclusive when $f'(x^*) = 0$.

Consider (i) $\dot{x} = -x^3$, (ii) $\dot{x} = x^3$, (iii) $\dot{x} = x^2$,

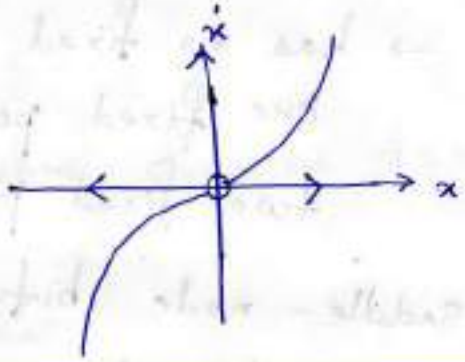
Above examples all have $f'(x^*) = 0$ for $x^* = 0$.

Graphical analysis:-

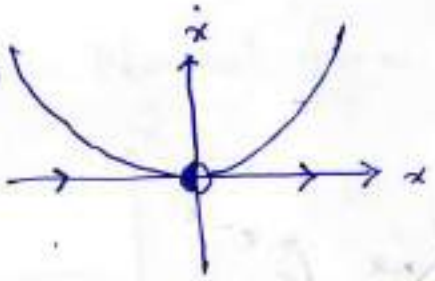
(i)



(ii)



(iii)



~~POTENTIAL PICTURE.~~

~~– Visualize a particle sliding down a potential well:-~~

~~$f(x) = -\frac{dV}{dx}$~~

~~$\frac{dV}{dt} = \frac{dV}{dx}\dot{x}$~~

~~$= -\left(\frac{dV}{dx}\right)^2 \leq 0$.~~

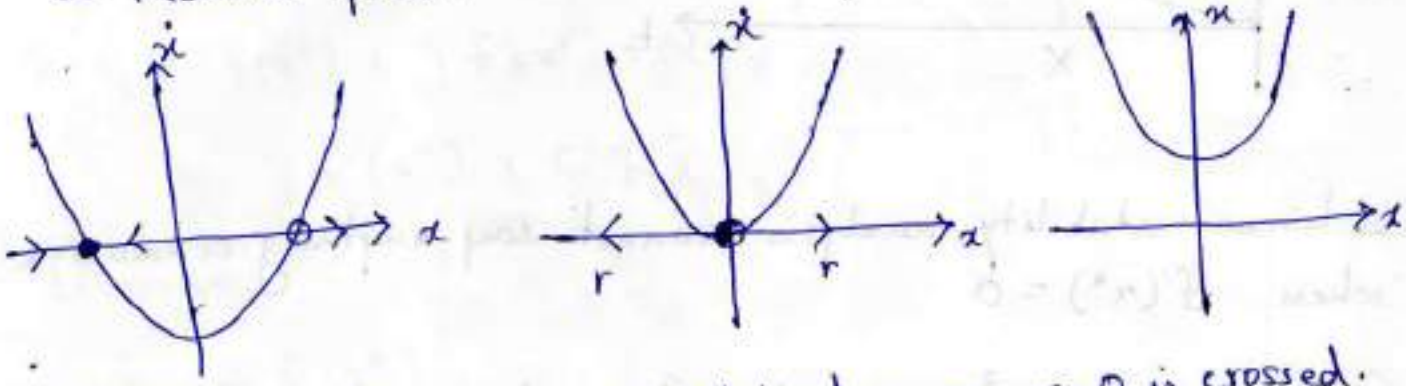
~~$\Rightarrow V(t)$ decreases along trajectories except if $\frac{dV}{dx} = 0$ (equilibrium)~~

~~So local minima of $V(x)$ $\rightarrow$ corresponds to stable fixed points.~~

BIFURCATIONS

– Saddle-Node Bifurcation.

– Normal form:- $\dot{x} = r + x^2$, r can be $\pm$ve or 0.



– Situation changes qualitatively as $r=0$ is crossed.

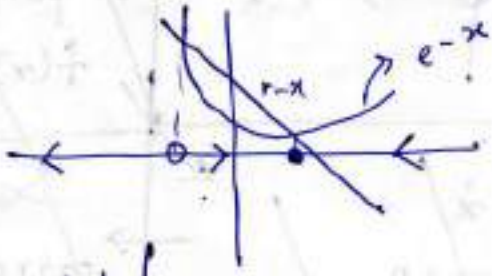
– BIFURCATION.

$\dot{x} = r - x^2$ → has no fixed points for $r<0$,
one fixed point for $r=0$.
two fixed points for $r>0$.

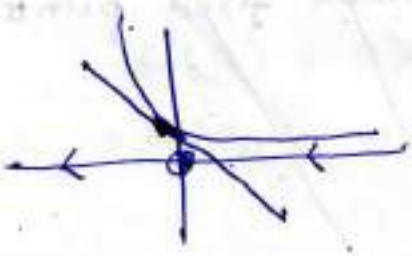
– In saddle-node bifurcation, a fixed point is created or destroyed.

Example:- $\dot{x} = r - x - e^{-x}$.

$f(x^*) = r - x^* - e^{-x^*}$



Consider $(r-x)$ and e^{-x}.



To find bifurcation point r_c, impose the condition that $(r-x)$ and e^{-x} intersect tangentially.

$$e^{-x} = r - x$$

$$\frac{d}{dx}(e^{-x}) = \frac{d}{dr}(r-x)$$

$$\Rightarrow \quad e^{-x} = 1$$

$$\Rightarrow \quad x = 0$$

$\Rightarrow \quad r_c = 1.$

and it occurs at $x = 0$.

Also, $\dot{x} = r - x - e^{-x}$

Expand near $x = 0$,

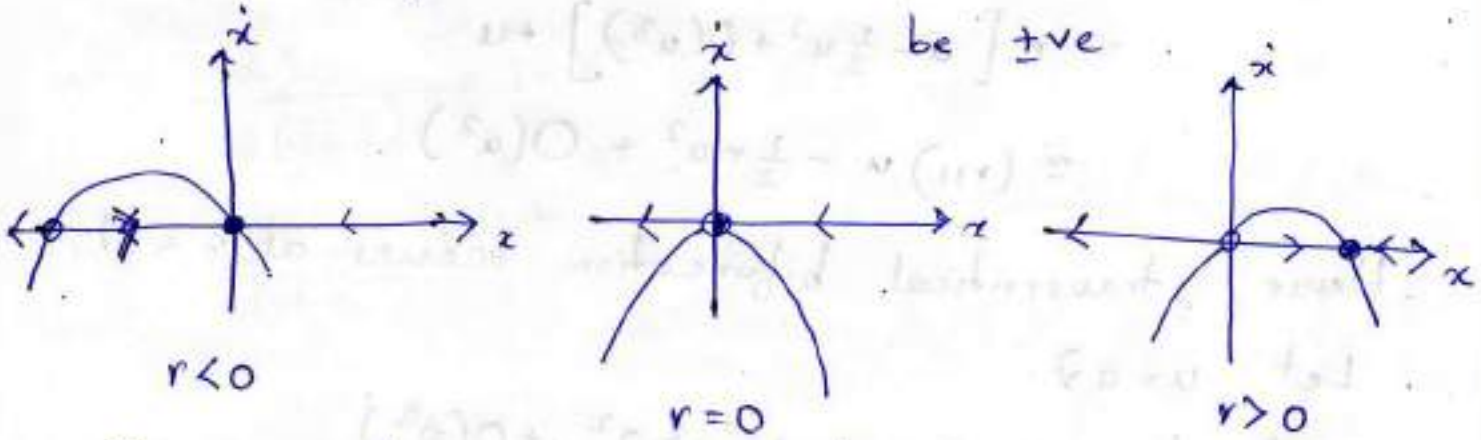
$$\dot{x} \simeq (r-x) - \left[1 - x + \frac{x^2}{2!} + \cdots\right]$$

$$\simeq (r-1) - \frac{x^2}{2!} + \cdots\cdots$$

$\hookrightarrow$ same algebraic form as $\dot{x} = r - x^2$.

— Transcritical Bifurcation:

— Normal form:- $\dot{x} = r\dot{x} - x^2$, allow x and r to be $\pm$ve



— Fixed point is not created/destroyed, but stability changes.

Example:- $\dot{x} = x(1-x^2) - a(1-e^{-bx})$.

$\dot{x} = 0 \Rightarrow x(1-x^2) - a(1-e^{-bx}) = 0 \rightarrow$ satisfied for $x = 0$. $\forall\ (a,b)$

$$1 - e^{-bx} = 1 - \left[1 - bx + \tfrac{1}{2}b^2x^2 + O(x^3)\right]$$

$$= bx - \tfrac{1}{2}b^2x^2 + O(x^3).$$

$$\therefore \dot{x} \simeq x - a\left(bx - \tfrac{1}{2}b^2x^2\right) + O(x^3)$$

$$= (1-ab)x + \tfrac{1}{2}ab^2x^2 + O(x^3)$$ $\rightarrow$ resembles normal form for transcritical bifurcation.

Example:- $\dot{x} = r\ln x + x - 1$.

$\dot{x} = 0 = r\ln x + x - 1 \rightarrow$ satisfied for $x = 1\ \forall\ r$.

Define $u = x - 1$

$$\dot{u} = \dot{x} = r\ln(1+u) + u$$

$$= r\left[u - \tfrac{1}{2}u^2 + O(u^3)\right] + u$$

$$\simeq (r+1)u - \tfrac{1}{2}ru^2 + O(u^3).$$

Hence, transcritical bifurcation occurs at $r_c = -1$

Let $u = a\nu$.

$$a\dot{\nu} = a(r+1)\nu - \tfrac{1}{2}ra^2\nu^2 + O(\nu^3).$$

$$\Rightarrow \dot{\nu} = (r+1)\nu - \tfrac{1}{2}ra\nu^2 + O(\nu^3).$$

Choose $R = r+1$ and $X = v$.

$$\dot{X} \approx RX - X^3 + O(X^3).$$

$$X = v = \frac{u}{a} = \frac{1}{2} r(x-1).$$

~~Solution~~

$$\dot{x} = a - bx^2 \quad , \quad a, b > 0.$$

~~Rescale,~~ $\cancel{\dfrac{dx}{d(at)} = 1 - \dfrac{x^2}{(a/b)}}$.

~~Let~~ $\cancel{X = \dfrac{x}{k^*}}$, ~~where~~ $\cancel{k^* = a/b}$, $\cancel{T' = \sqrt{ab}\,t}$

$$\cancel{k \frac{dX}{d(at)} = 1 - X^2}. \qquad \dot{x} = a\left(1 - \frac{x^2}{a/b}\right)$$

$$\cancel{\Rightarrow \frac{dX}{}} \qquad \Rightarrow \frac{dx}{d(at)} = 1 - \frac{x^2}{a/b}$$

$$\Rightarrow \frac{dx}{d(at)} = 1 - X^2. \qquad X = \frac{x}{\sqrt{a/b}}$$

$$\Rightarrow \sqrt{\frac{a}{b}} \frac{dX}{d(at)} = 1 - X^2.$$

$$\Rightarrow \frac{dX}{d(\sqrt{ab}\,t)} = 1 - X^2$$

$$\Rightarrow \frac{dX}{dT} = 1 - X^2$$